Finite-population correction for confidence intervals

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Finite N, sample without replacement, and n/N>0.05.
- Need a CI for μ or π.
- Sampling a material fraction reduces uncertainty.

2 DECISION / METHOD

1. Build the ordinary SE.
2. FPC=sqrt((N-n)/(N-1)).
3. adjusted SE=ordinary SE*FPC.
4. Use the usual Z/t critical and centre.
5. No FPC for replacement or negligible fraction.

3 FORMULA

FPC=√[(N-n)/(N-1)]
mean CI: $\bar{X} \pm$ critical×(s/√n)×FPC
proportion CI: p ± z*√[p(1-p)/n]×FPC

4 FORMULA SHEET VS WHAT I KNOW

Gives: finite-population sample-size adjustment only.
Still know: CI FPC multiplier, trigger, and replacement caveat.

5 EXCEL ROUTE

- FPC=SQRT((N-n)/(N-1))
- multiply ordinary SE by FPC before the margin.
- retain Z/t choice from the underlying CI method.

6 TINY WORKED EXAMPLE

N=1000,n=100,p=.30,C=.90: FPC=.9492; adjusted SE=.0435. Using full precision, margin=.0715 and CI≈(.2285,.3715).

7 TOP EXAM TRAPS

- Applying FPC to the centre.
- Dividing by FPC.
- Applying it when sampling with replacement.
- Forgetting the n/N trigger.

8 SAY ALOUD

For material sampling without replacement, FPC multiplies the ordinary standard error and narrows the interval.

9 EXIT CHECK

Where in the CI formula does FPC belong?

ONE-GLANCE VISUAL ROUTE

